

Water Quality Event Detection in Drinking Water Network

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Abstract Drinking water network is vulnerable to toxic chemicals. Anomaly detection-based event detection can provide reliable indication of contamination by analyzing the real-time water quality data, collected by online-distributed sensors in water network. This article reviews the water quality event detection methodologies based on the correlation of water quality parameters and contaminants. Further, we review how to reduce the impact of contamination in water distribution network, including sensor placement optimization and contamination source determination.

Keywords Drinking water · Contamination · Event detection · Free chlorine · Impact reduction

1 Introduction

Drinking water network is vulnerable to contaminant intrusion in a terrorist attack or other pollution scenarios (Zhang et al. 2010a, b; Middelkoop 2000). So, we are trying to use real-time water quality data to provide warning to contamination events to safeguard the water. Water quality is the physical, chemical, and biological characteristics of water. Since the 1990s, automated sampling, analyzing, and communication techniques have been implemented in real-time water quality monitoring (Cook et al. 2006). So, a contamination warning system can be set up to monitor the water quality and provide warning to contamination by analyzing the real-time water quality data. Many countries like the USA, China, Germany, and Japan have deployed contamination warning systems (CWS) (Hou et al. 2013c).

Contamination warning systems provides an integrated approach to detect and respond to contamination events, which uses advanced monitoring technologies and enhanced surveillance activities to collect, integrate, analyze, and communicate information and provide timely warnings of potential contamination incidents in order to make rapid and effective response actions (Koch and McKenna 2010; USEPA 2005b, 2010). CWS integrates online monitoring system, supervisory control and data acquisition (SCADA), event detection system, and decision support system (USEPA 2010), which implements water quality data acquisition, transmission, storage,

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Fig. 1 The structure of a contamination warning system. The four components are as follows: online monitoring system, supervisory control and data acquisition, event detection system, and decision support system

and analysis to provide water quality alarms and suggestions for rapid response, as shown in Fig. 1. Sensor data is collected from the online monitoring system. Then, the data is formatted and transferred to event detection system through SCADA. In event detection system event, detection result is obtained by analyzing the data. Decision support system provides support information by analyzing the event detection results.

Event detection system in contamination warning system can provide indications of potential contamination by analyzing real-time water quality data (USEPA 2005b, 2010). Event detection provides indications of abnormal water condition, such as contamination events or other abnormal condition.

The conventional water quality parameters can be used as surrogate parameters for the detection of toxic chemicals. Experiments show the correlation between some water quality parameters and toxic chemicals (USEPA 2005a; Jeffrey Yang et al. 2009; Byer and Carlson 2005). Water quality data change significantly when certain contaminants are introduced into water. Many event detection methodologies have been proposed. The methodologies mainly consist of the sensor data analysis, event detection methods, and performance tests. Sensor data analysis is to reduce the impact of background variation, instrument noise, and drifting in water quality data, and make the water quality data aligned. Event detection methods is to analyze the real-time water quality data by comparing the new data pattern with the normal data pattern, using statistical properties, data mining methods, and time-frequency analysis. Performance test is mainly to verify the detection methods to meet the requirements of accuracy.

Further, when contamination event occurs, faster detection and response is important, including to determine the contamination source and minimize the population affected. Contamination source determination and sensor placement optimization methods are developed to find the contamination source and minimize the impact.

In this review, we focus on the water quality event detection and impact reduction methods. In the second

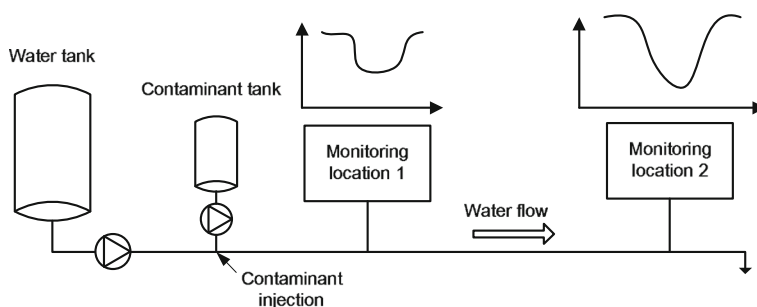
section, the laboratory experimental results for the correlation between water quality parameters and toxic chemicals are presented. In the third section, water quality event detection methodologies are reviewed, including the sensor data analysis, event detection methods, and performance test. Sensor placement optimization and contamination source determination methods are reviewed for impact reduction. Finally, we discuss the research needs.

2 Correlation and Modeling of Water Quality Parameters and Contaminants

Water contaminants can be divided into three categories: chemical, microbial, and radiological (USEPA 2011). Because so many potential contaminants exists, real-time monitoring for every potential contaminants is impractical for the huge cost and operational difficulties. So, to use conventional water quality parameters as surrogate parameters to provide indications of contaminants is efficient for contamination warning (Ziglio and Hunsinger 2002). Conventional water quality parameters mainly include free chlorine, total chlorine, chloride, pH, turbidity, conductivity, ammonia-nitrogen, nitrate-nitrogen, total organic carbon (TOC), and oxidation-reduction potential (ORP). In practical situations, some parameters are widely monitored such as the chlorine and turbidity in drinking water network, while some parameters are not widely monitored, such as TOC.

Contaminants Introducing Experiments Contaminants introducing experiments are carried out in the laboratory to find the correlation between contaminants and conventional parameters (USEPA 2005a; Jeffrey Yang et al. 2009; Byer and Carlson 2005). Toxic chemicals are injected into water pipes simulating the chemicals injected into water network, and water quality are measured by online monitoring instruments before and after injection. Experiments can be conducted in recirculating pipe-loop mode or single-pass pipe mode. Recirculating mode can

Fig. 2 A typical contaminant injection experiment pipe system. The water flows from the water tank, then passes the contaminant injection point and monitoring locations. Above the monitoring locations are possible data curves for free chlorine



simulate longer flow of contaminants (USEPA 2005a). A typical experiment system is shown in Fig. 2, in which the two curves above the monitoring locations are possibly free chlorine monitoring data curves when contaminant is introduced. The drop of free chlorine data at the second monitoring location is more significant than at the first one, showing that free chlorine is consumed when water flows from the first monitoring location to the second one.

Special chemicals are selected for experiments. Pesticide, herbicides, alkaloid, bacterium, and biological growth media and inorganic laboratory chemicals are the mostly used contaminants in laboratory injection experiments (Jeffrey Yang et al. 2009; USEPA 2005a; Kroll and King 2006). High toxicity, high water solubility, chemical and physical stability, a low chance of detection, and a lack of taste, color and odor are the key properties for identifying credible threat contaminants (Byer and Carlson 2005; Deininger et al. 2000), which can also be key properties of chemicals selected for experiments. Though quite a few chemicals are selected for injection experiments, a broader range for chemicals is required for comprehensive study.

Parameter Change Patterns Injection experimental results show that certain water quality parameters change rapidly in a short time frame when chemicals are introduced into water (Grayman et al. 1988). Parameters changes 10 to 50 or over 50 % from normal water quality condition in response to different contaminants. Some of the parameters change significantly, others change a little, or no change occurs. The parameters which change significantly are considered to be sensitive to these chemicals, and the parameters which change a little or with no change is considered to be insensitive. Experiment results show that free chlorine, and TOC are the most sensitive parameters (USEPA 2005a; Hall

et al. 2007), which change significantly when most of the contaminants are introduced into water. And the free chlorine is considered to be the most useful parameter, because it widely exists in drinking water as the disinfectant and can react with many contaminants.

For the sensitive parameters, different change patterns exists when contaminants are introduced into water. These change patterns can be used to discriminate the contaminants. The patterns, called “fingerprints,” can be obtained using the time series of the distance between the current parameter vector and the baseline (King and Kroll 2005). The minimum detection limits experiments show that many contaminants can be detected below 5 % of LD50 using fingerprints (in 1 L of water). Concentration-time change patterns of water quality parameter can be classified into U-shaped loss pattern, inverse U-shaped pattern, and V-shaped pattern (Jeffrey Yang et al. 2009).

Water quality parameters also correlated with bacterial contaminants. *Escherichia coli* is mostly used for laboratory experiments as a representative of bacterial contaminants (Helbling and VanBriesen 2007, 2008; USEPA 2005a; Jeffrey Yang et al. 2009). Results show detectable free chlorine changes when *E. coli* is introduced into water. The change magnitude of chlorine residual is proportional to cell concentration, background free chlorine concentration, and contact time (Helbling and VanBriesen 2008). Experiment results show that the chlorine demand of microbial suspensions is species specific. Experiment results show that chlorine-resistant organisms also demand chlorine (Helbling and VanBriesen 2007).

A comparison of water quality parameter changes in contaminant-introducing experiments is presented in Table 1. The change patterns of water quality parameters are substantially consistent in different experiments.

Table 1 Comparison of water quality parameter changes in contaminant-introducing experiments

Contaminant	Cl_f	Cl_t	Chloride	pH	ORP	TOC	Cond.	Tur.	DO	AN	NN	Reference
Aldicarb	L	L	/	/	/	/	/	/	S	M	/	(USEPA 2005a)
	–	–	NC	NC	–	+	NC	+	NC	NC	NC	(Hall et al. 2007)
	–	/	/	NC	/	+	NC	+	/	/	/	(Kroll and King 2006)
	L	/	/	L	/	M	S	M	/	/	/	(Byer and Carlson 2005)
	L–	L–	S+	NC	S–	/	/	/	/	/	/	(Jeffrey Yang et al. 2009)
Nicotine	L	M	/	/	/	/	/	/	S	M	/	(USEPA 2005a)
	–	NC	+	NC	–	+	NC	NC	NC	+	–	(Hall et al. 2007)
	–	/	/	NC	/	+	NC	+	/	/	/	(Kroll and King 2006)
	M–	M–	NC	NC	S–	/	/	/	/	/	/	(Jeffrey Yang et al. 2009)
Potassium ferricyanide	S	M	/	/	/	/	/	/	S	S	/	(USEPA 2005a)
	–	NC	+	NC	–	+	+	+	NC	NC	NC	(Hall et al. 2007)
	NC	M+	M+	NC	NC	/	/	/	/	/	/	(Jeffrey Yang et al. 2009)
Glyphosate	L	L	/	/	/	/	/	/	S	M	/	(USEPA 2005a)
	–	NC	+	NC	–	+	NC	NC	NC	NC	NC	(Hall et al. 2007)
	L–	L–	S+	S–	S–	/	/	/	/	/	/	(Jeffrey Yang et al. 2009)
Arsenic trioxide	–	–	NC	NC	–	NC	NC	+	NC	+	–	(Hall et al. 2007)
	–	/	/	+	/	NC	+	+	/	/	/	(Kroll and King 2006)
<i>Escherichia coli</i>	–	–	NC	NC	NC	+	NC	+	NC	+	NC	(Hall et al. 2007)
	–	/	/	–	/	+	+	+	/	/	/	(Kroll and King 2006)
	–	/	/	/	/	/	/	/	/	/	/	(Helbling and VanBriesen 2008)
	L–	S–	S+	NC	S–	/	/	/	/	/	/	(Jeffrey Yang et al. 2009)
	L	M	/	/	S	/	/	/	/	/	/	(USEPA 2005a)
Nutrient broth	S–	S–	NC	NC	S–	/	/	/	/	/	/	(Jeffrey Yang et al. 2009)
	S	S	/	/	S	/	/	/	/	/	/	(USEPA 2005a)
Terrific broth	M–	S–	NC	NC	S–	/	/	/	/	/	/	(Jeffrey Yang et al. 2009)
	M	S	/	/	S	/	/	/	/	/	/	(USEPA 2005a)
Tryptic soy broth	S–	S–	NC	NC	S–	/	/	/	/	/	/	(Jeffrey Yang et al. 2009)
	M	M	/	/	S	/	/	/	/	/	/	(USEPA 2005a)
Dimethyl sulfoxide	M	M	/	/	/	/	/	/	S	S	/	(USEPA 2005a)
Sodium thiosulfate	L	L	/	/	/	/	/	/	S	M	/	(USEPA 2005a)
Waste water	+	NC	+	NC	+	+	NC	NC	NC	+	+	(Hall et al. 2007)
Malathion	–	NC	NC	NC	–	+	NC	+	NC	NC	NC	(Hall et al. 2007)
Terrific broth	–	–	NC	NC	–	+	NC	+	NC	NC	NC	(Hall et al. 2007)
Sodium arsenate	/	S	/	S	/	S	M	L	/	/	/	(Byer and Carlson 2005)
Sodium cyanide	L	/	/	M	/	S	M	L	/	/	/	(Byer and Carlson 2005)
Sodium fluoroacetate	S	/	/	M	/	S	S	M	/	/	/	(Byer and Carlson 2005)

Table 1 (continued)

Contaminant	Cl_f	Cl_t	Chloride	pH	ORP	TOC	Cond.	Tur.	DO	AN	NN	Reference
Dicamba	NC	NC	NC	S–	S+	/	/	/	/	/	/	(Jeffrey Yang et al. 2009)
Mercuric chloride	NC	NC	NC	S+	S–	/	/	/	/	/	/	(Jeffrey Yang et al. 2009)
Colchicine	L–	L–	S+	NC	S–	/	/	/	/	/	/	(Jeffrey Yang et al. 2009)

Cl_f is free chlorine, Cl_t is total chlorine, *Cond.* is conductivity, *Tur.* is turbidity, *AN* is ammonia-nitrogen, *NN* is nitrate-nitrogen, *L* is large changes, *M* is moderate changes, *S* is small changes or no change, */* stands for not measured parameters in experiments, *–* is decrease from the baseline, *+* is increase from the baseline, *NC* is no change from the baseline, *L–* is large decrease from the baseline, and so on

Modeling of Contaminants Reactivity and Transport

As free chlorine is the most sensitive and widely monitored parameter in drinking water as an indication of contaminants, the modeling of chlorine and contaminant reaction is meaningful. Many models for chlorine decay have been developed (Haas and Karra 1984), including the first order, second order, *n*th order, parallel first order, limited first order, and limited *n*th order models. Different kinetic models for chlorine decay are tested, using bottle tests (Powell et al. 2000) and microbial injection tests (Helbling and VanBriesen 2009).

Competitive second-order kinetic model (Clark and Sivaganesan 2002) is applied to obtain theoretical chlorine loss in a contaminant “slug” (Jeffrey Yang et al. 2009). Contaminants are classified into fast-reacting, slow-reacting, and non-reacting types according to different reaction kinetics. The fast-reacting contaminants have large reaction kinetics, the slow-reacting contaminants have small reaction kinetics, and the non-reacting contaminants have nearly zero reaction kinetics, and the reaction kinetics are estimated using experiment results (Yang et al. 2007). Models of chlorine decay are also applied to the reactivity data of bacterial contaminants and free chlorine. The parallel first-order model provides the best fit of the experimental chlorine decay data generated by dense microbial suspensions (Helbling and VanBriesen 2009). Another biphasic exponential decay model is used to analyze the chlorine demand curves of microorganisms (Helbling and VanBriesen 2007).

When contaminants invade into the drinking water network, the contaminants will move with the drinking water. Transit time is the time that contaminants transfer from one location to another, which can be

estimated using temperature or conductivity pulses introduced into a pipe (O Halloran et al. 2006). Transit time vary in the range of a low value in the afternoon to a high value before dawn due to different water demand. Mathematics model of contaminants transport in drinking water pipes is also studied (Jeffrey Yang et al. 2008). When introduced into water, fast-reactive contaminants form a slug and react with free chlorine quickly, and a reactive interface is formed between the water and contaminants. Longitudinal dispersion is the controlling factor in non-reactive contaminants transport to form the slug, and the slow-reacting contaminants follow compromise controlling factors.

3 Water Quality Event Detection Methodologies

Based on correlation of contaminants and water quality parameters, the changes of water quality parameters can be used to indicate potential contaminants. Anomaly detection methods are applied to detect the changes of parameters. Anomaly detection methods have been previously developed and applied in network intrusion detection, fraud detection, medical and public health anomaly detection, industrial damage detection, and many other fields (Chandola et al. 2009). Water quality event detection methodologies include the sensor data analysis, anomaly detection methods, and performance tests, as shown in Fig. 3.

Some basic definition is as follows:

Definition 1 (Outlier) (or called anomaly) For a series of water quality data $x(k)$, outlier is the data point which is not consistent to the normal pattern.

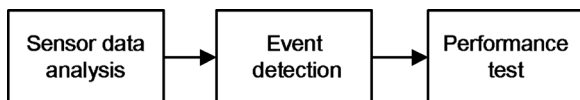


Fig. 3 Three components of event detection

Definition 2 (Event) Event is a series of continuous significant changes, which is possibly caused by contamination events.

Definition 3 (Baseline) Baseline is the stable water quality in the normal condition.

Definition 4 (Baseline change) Baseline change is the persistent changes in mean of water quality parameters, usually due to network operations.

Typical baseline change, outlier, and event in real-time monitoring data are represented in Fig. 4.

Sensor Data Analysis Sensor data analysis mainly reduces the impact of noise and make the data more suitable for further analysis. Water quality data normalization is a necessary process because different water quality parameters have magnitudes in different units, which may lead to biased detection result. Data with different units is usually normalized to a

common scale by subtracting the data set mean μ and then dividing the difference by the standard deviation. In another normalization method, the data is subtracted from the minimum value of the data and then divided by the difference between the maximum and minimum value of the data.

$$x(t) = \frac{\dot{x}(t) - \mu}{\sigma} \quad (1)$$

$$x(t) = \frac{\dot{x}(t) - \min(\dot{x}(t))}{\max(\dot{x}(t)) - \min(\dot{x}(t))} \quad (2)$$

$\dot{x}(t)$ is the original water quality data in the t time step, and $x(t)$ is the normalized water quality data.

Water quality signals consist of signals responding to contaminant, background signal from natural variation instrument noise and drifting, and other noise. Anomaly detection methods are sought to identify the contaminant-responding signals in form of water quality changes and then separate it from background variations (Jeffrey Yang et al. 2009). So, effective de-noising is helpful for better detection result. A robust statistical method is proposed to track and characterize natural background variations and instrument drift, using a time-series polynomial function (Yang et al. 2007; Jeffrey Yang et al. 2009). Transformation of the original water quality data is used. A differencing filter is used to calculate the first derivative of the water quality data series which

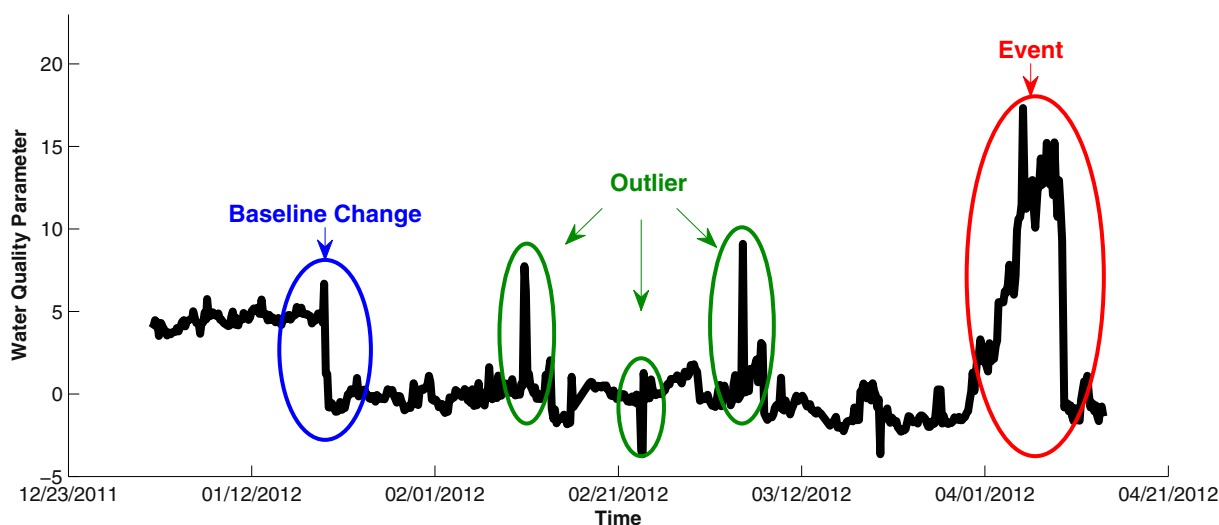


Fig. 4 Example of baseline change, outlier, and event. The curve is some water quality parameter in about 3 months. Baseline change is persistent changes in the mean of water quality

data. Outlier is a stand-alone point which is not consistent to the normal pattern. Event is a series of continuous significant changes

can be used to find the increasing or decreasing trends (Murray et al. 2011). The original water quality data also can be transformed into binary data (Kumar et al. 2012).

Water quality signals consist of instrument error data, which may significantly reduce the performance of event detection. Data smoothing method using local polynomial regression fitting is applied to remove noise and error data (Liu et al. 2013). Wavelet analysis is also applied to remove noise and error data (Hou et al. 2013a), which decomposes the water quality time series data into several components and threshold is applied to the components to remove the noise or error data.

The baseline characterizes the normal water quality condition (Byer and Carlson 2005). The deviations from baseline should be evaluated to determine whether it is normal variation or the indication of contaminants (USEPA 2003). Baseline can be obtained using sliding window for water quality time series (Byer and Carlson 2005). Baseline change is persistent water quality change due to network operation (McKenna et al. 2007). A composite signal method is proposed to distinguish the changes caused by contamination or operation, by incorporating the signals of network objects (Hart et al. 2010).

Water Quality Event Detection Methodologies and Scheme Water quality event detection compares the new data pattern with the normal data patterns for event detection. If the new data pattern is different from the normal data pattern, then it is considered to be an anomaly. An anomaly score is usually obtained for identifying anomaly, which is compared with a normal threshold.

Definition 5 (Anomaly score) Anomaly score is used to discriminate the outlier by comparing to a threshold.

If the anomaly score exceeds the threshold, then the corresponding data can be declared to be an anomaly. Then, a series of outliers can be discriminated as an event, as shown in Fig. 5.

Several water quality change types have been proposed (USEPA 2010), including the statistical properties-based methods, data mining methods, and distributed sensor fusion methods. In statistical properties-based method, water quality data is

assumed to follow normal distribution, and 99.96 % of the points should fall within the three standard deviations from the mean ($\bar{x} \pm 3\sigma$) (Byer and Carlson 2005). $\bar{x}$ is the mean of the water quality data and the σ is the standard deviation. The data, which is a three-fold standard deviation away from the normal mean, is declared as an anomaly. So, the anomaly score is calculated as:

$$a(t) = x(t) - \bar{x} \quad (3)$$

$a(t)$ is the anomaly score in the t time step. In time series increments method, the anomaly score is simply calculated as:

$$a(t) = x(t+1) - x(t) \quad (4)$$

In linear predictor method, the water quality parameter is predicted for the next time step, and the predicted value is compared to the measured value. The residual is used as anomaly score (McKenna et al. 2008):

$$a(t) = r(t) = \hat{x}(t) - x(t) \quad (5)$$

the $\hat{x}(t)$ is the predicted value and $r(t)$ is the residual in the t time stamp. It is similar in artificial neural network-based water quality event detection methods (Perelman et al. 2012; Hou et al. 2013a), which use artificial neural network to predict the water quality data. In a dynamic threshold-based method, the anomaly score is just the water quality data itself, but the threshold is adaptive according to the current water quality (Arad et al. 2013):

$$a(t) = x(t) \quad (6)$$

In data mining-based methods, clustering is widely applied for water quality event detection. In clustering-based method, the water quality is treated as a vector:

$$X(t) = [x_1(t), x_2(t), \dots, x_n(t)] \quad (7)$$

where $x_i(t)$ is the i th parameter at t time step.

Water quality event detection based on K -means clustering (Klise and McKenna 2006), fuzzy c -mean clustering (He et al. 2013), and other clustering method (Raciti et al. 2012) are applied to the water quality data set, and the anomaly score is calculated

as the distance from the data point to the clustering center:

$$a(t) = \text{dist}(X(t) - C(i)) \quad (8)$$

dist is the distance function for two data point, and $C(i)$ is the clustering center.

Performance test for time series increment, linear predictor, and clustering-based methods show that the performance changes according to the data set (McKenna et al. 2006). Time-frequency analysis is effective for water quality event detection. Time-frequency analysis allows the decomposition of a signal into individual frequency components and establishes the relative intensity of each component (Cohen 1989), and the decomposition coefficients are used as anomaly score. A empirical mode decomposition-based method (Liu et al. 2013) is proposed for water quality event detection, which uses the local time scale as the anomaly score. Events can be discriminated based on anomaly detection results for a series of single point. Event discrimination methods using binomial event discriminator (BED) (McKenna et al. 2007) and Bayesian probability (Arad et al. 2013) are proposed.

Several water quality parameters may change simultaneously when contaminants are introduced into water. So, the fusion for the event detection results of a single parameter may provide a more reliable result. Fusion is particularly useful in drinking water network where many sensors are deployed. Simulation of network fusion show that fusion approach can reduce the false alarm rate by three orders of magnitude (Koch and McKenna 2010). Fusion

method is proposed based on extended Dempster-Shafer method (Hou et al. 2013b) and fuzzy logic-based method (Liu et al. 2013). In the Dempster-Shafer method, the probability of event is calculated combining each single probability obtained by a single parameter:

$$m(A) = m_1 \oplus m_2 \oplus \cdots \oplus m_n(A) \quad (9)$$

The $m_i(A)$ is the probability function for the i th parameter, $m(A)$ is the fused probability and $\oplus$ denote the orthogonal sum of two probability.

Distribution Network-Based Water Quality Event Detection Water distribution network consists of pipes, pumps, valves, fittings, and storage facilities that convey water from source to consumers (Shang et al. 2007). Active pumps and pump flows, alarms, valve status, and pressure all can be used in water quality event detection for feature selection and clustering (Raciti et al. 2012).

Operations of objects in distribution network may cause significant changes of water quality, which can cause false alarms in event detection. Composite signal with tank level information and conductivity data is used as calibration information to reduce false alarms (Hart et al. 2010). Temporal and space correlation in distribution network is used to reduce false background alarms (Koch and McKenna 2010). A space-time zone is used to detect significant clusters of events in distribution network using Kull-dorff's test, utilizing the travel time between sensing nodes. Bayesian belief network-based event detection method is proposed, based on the causal relationships of contamination and water quality data (Dawsey et al. 2006; Murray et al. 2011).

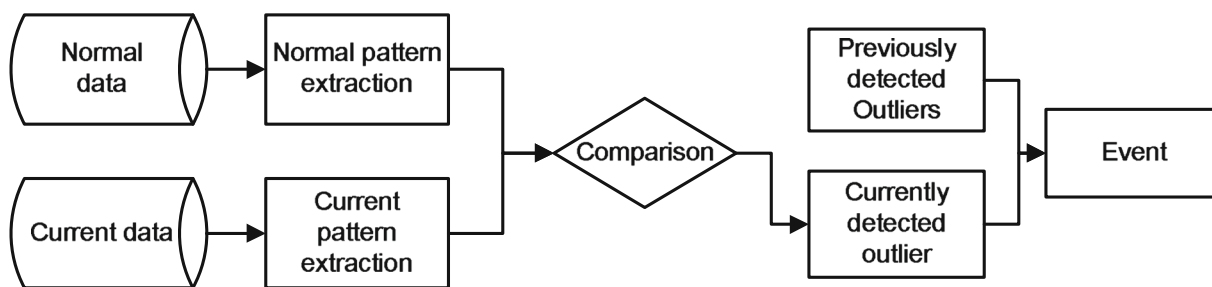


Fig. 5 Event detection flow chart. Firstly, normal pattern is extracted from normal data. Then, current pattern is extracted from current data and compared with the normal one. If the

current pattern is significantly different from the normal one, then the current data is declared as an outlier. Series of outliers are declared as event

As a sensitive parameter, free chlorine is the most important parameter in water network. Due to the decay of chlorine and the supply of fresh water, the concentration of free chlorine fluctuates in a certain range. Based on the upper and lower bounds of the free chlorine, the data points that exceed the upper and lower bounds can be declared as anomaly (Eliades and Polycarpou 2013). The upper and lower bounds are computed based on water demand patterns and reaction coefficients of chlorine.

Simulation Tools and Experiments Simulation tools are developed to simulate transport and reactivity of contaminants in a distribution network. EPANET and EPANET-MSX are widely used for simulation of chlorine and contaminant dynamics (Shang et al. 2006; Shang et al. 2008; Eliades and Polycarpou 2013; Helbling and VanBriesen 2009, verification of distributed sensor fusion method, and contaminant source characterization method in network (Koch and McKenna 2010; Kumar et al. 2012). EPANET takes a description of distribution network, and simulates the network model, hydraulics, contamination injection, water demands, and contaminants transport. EPANET's water quality simulator uses a Lagrangian time-based approach to track the fate of discrete parcels of water as they move along pipes (Rossman 2000).

Performance Test The performance test is to evaluate the performance of event detection methods. The accuracy and the detection latency time are the most important factors for performance. The higher accuracy means detecting more events and less false alarms. The accuracy can be assessed using the area under receiver operating characteristic (ROC) curve (McKenna et al. 2006; Kroll and King 2010). The ROC curve uses the probability of detection (PD), false alarm rate (FAR), and other parameters. Probability of detection and false alarm rate can be calculated as:

$$PD = \frac{TP}{TP + FN}, FAR = \frac{FP}{FP + TN} \quad (10)$$

True positive (TP) and true negative (TN) are cases that reported alarms are consistent with the event occurrence, which means alarms are reported or not meanwhile event occurs or not. False positive (FP) or false negative (FN) is the case that alarms are not

consistent with the event occurrence, which means alarms are reported with no event or alarm is not reported when event actually occurs. Detection latency time is also important. Latency time is suggested to be counted from the time when the peak of the event has taken place (Raciti et al. 2012).

To overcome the difficulties of lack of contamination data, simulated events are used. Simulated events with different strengths and duration are superimposed on the baseline water quality for simulation of contamination (McKenna et al. 2006; USEPA 2013). Further, contaminants concentration profiles are used for the simulation of real injection of contaminants (Raciti et al. 2012).

A comprehensive and detailed performance test called event detection system challenge is carried out for available EDSs which is implemented under the US Environmental Protection Agency (EPA) Water Security initiative (USEPA 2013). The available EDS participated include CANARY developed by Sandia National Laboratories, ana::tool by s::can, OptiEDS by OptiWater (Elad Salomons), Event Monitor by Hach Company, and BlueBox™ by Whitewater Security. The challenge uses 1-year long monitoring data from six monitoring station from four US water utilities, with the first 3-months data used for training and the remaining for testing which is called baseline data. Baseline events, which is embedded in the original data, and simulated events, which is superimposed on the baseline data, are used to test the EDSs' detection performance. Firstly, the challenge analyzes the invalid and valid alerts generated by EDSs, which are corresponding to false and true alarms, and invalid alerts are assigned to one of the causes including normal variability, sensor problem, data communication, and no clear cause. Result shows that normal variability is the main cause for invalid alerts and alert causes are clearly station dependent, and detection rate are fairly consistent across the monitoring station. Then, the challenge analyzes the detection performance under different alert threshold, which shows the tradeoff between valid alerts and invalid alerts are useful for water utilities to adjust the parameters, and result shows that no EDS has the best performance universally. Thirdly, all factors impacting the detection rate is tested including the six monitoring locations, six types of contaminants and high or low concentration, STEEP or FLAT event profile, and different event start time, and result shows that the event start

time has the biggest influence on the detection performance for all the EDS, and each EDS has at least one factor that highly impacted its performance over 25 %. Several suggestions on how to select EDS for water utilities are proposed after the challenge. And several valuable opinions are proposed that traditional ROC curve uses pointwise false alarm calculation is not valid for EDS comparison for the difficulty in identifying the “sample,” and the area under a ROC curve may be misleading in EDS evaluation.

4 Impact Reduction for Contamination Events

In drinking water network, water quality sensors are distributed in the network providing real-time water quality data. A better sensor placement can help to detect contamination quickly and robustly. And contamination source determination methods using real-time water quality data can help to find the contamination source as quickly as possible and reduce the damage caused by contamination.

Sensor Placement Optimization Online sensors are a critical component of CWS, which collect real-time data for routine monitoring and contamination event detection. Since setting up sensors at each node in the network is prohibitively expensive, sensors have a limited number. So, optimal sensor placement to minimize the risks becomes a crucial issue and the key element of CWS design (Krause et al. 2008; Hart et al. 2007; Hart and Murray 2010). Sensor placement optimization method is automated for sensor placement search using computational methods which provides accurate design based on minimized risk (USEPA 2009).

Sensor placement optimization process is usually composed of a network modeling process and a decision-making process that employs optimization, which is studied in detail in the EPA's report in which TEVA-SPOT software is used for the optimization (USEPA 2009). Firstly, a water distribution network model is set up which is used to simulate the impacts of contaminant incidents using hydraulic network model to measure the effectiveness of sensor placement and provide suggestions on decision making. Steps to set up an accurate network model consist of network objects modeling, water demand

patterns determination, model calibration/validation, and testing (USEPA 2009).

Then, contaminant incidents are designed to simulate and calculate the impact of the contamination before or after the sensors installation to check the impact reduction to decide the sensor locations, sensor numbers, and the tradeoff between the impact reduction and the cost. Incidents are designed to occur at every possible node and every hour of the day for a comprehensive test, so the computation time and required memory for large network design is a challenge. Various performance objectives are designed to quantify the impact caused by the contamination, including expected time of detection, expected population affected prior to detection, expected consumption of contaminated water prior to detection, and detection likelihood which are objectives in the “Battle of the Water Sensor Networks” (Ostfeld et al. 2008). Other objectives includes number of people sickened after exposure, number of pipe feet contaminated, mass of contaminated water consumed (USEPA 2009) and detection redundancy (Preis and Ostfeld 2008). As there is a tradeoff between the objectives, the objectives can not be optimized at the same time. One way to solve the tradeoff optimization is considering one main objective first and then to check the rest of the objectives. Using this method, a “regret analysis” is proposed to measure the degree of optimization of a design, which is obtained by optimizing one main objective and compare the performance with optimal configuration checking the other objectives (USEPA 2009). Another way is to obtain a compromised non-dominated solution by optimizing a combined object using assigned weights for the objectives or using other fused forms (Dorini et al. 2010; Guan et al. 2006; Wu and Walski 2006). Another factor that greatly affect the incident impact is the response time which is the time to check the incident and warn the citizens not to use the water after a incident occurs, which should be taken into the impact estimation (USEPA 2009).

Once the objective is determined, a solver is needed for mathematical computation to find the best sensor locations. The solvers include the mixed-integer programming (MIP) (Berry et al. 2005), and heuristic search methods including Greedy Randomized Adaptive Search Procedure heuristic (GRASP) (USEPA 2009), population-based ant colony method (Diwold et al. 2010), and genetic algorithm-based methods (Ostfeld and Salomons 2005; Huang et al. 2006;

Weickgenannt et al. 2010). Greedy algorithm-based methods are also proposed (Krause et al. 2008) which places one sensor on each step to obtain the optimal solutions (Eliades and Polycarpou 2006), or in another method, initially sets a random sensor placement then replaces one sensor with another for maximum impact reduction in each step to obtain a final optimal placement (Dorini et al. 2010).

To limit the time and space requirement of the solvers is very important when optimizing large network design as it may take too much time or memory, so several optimization methods are proposed. Node aggregation is used for MIP solver to reduce the memory requirement and Lagrangian heuristic method reduces the memory used for GRASP (USEPA 2009). Pre-selection of candidate sensor locations and contamination scenarios can reduce the computational complexity (Diao and Rauch 2013; Huang et al. 2006; Weickgenannt et al. 2010).

Robust objective is proposed considering severe contamination incidents with worst-case performance (Krause and Guestrin 2009), as severe incidents cause great impact and the impact is averaged with other incidents in non-robust objective computation. There's a tradeoff between the average-case and worst-case optimization (Krause and Guestrin 2009). One method is to optimize the objectives with the maximum impact limited or to limit the impacts caused by a fraction of the incidents (USEPA 2009), and other methods are also proposed (Krause and Guestrin 2009; Ma et al. 2010).

Some other factors should be considered in sensor placement design; for example, some locations are more suitable for equipment installation and maintenance because of physical security, data transmission requirement, and other conditions, which should be important to candidate sensor locations such as the fire and police station or sewage systems sites (USEPA 2009).

Contamination Source Determination When intentional or accidental contamination event occurs, it is important to identify the time and location of contamination sources (Laird et al. 2005), which can be accomplished utilizing the results of contamination event detection as clues. Methods are developed to identify the contamination source.

The source determination problem is usually reformulated into an optimization problem, which is to

minimize the difference between measured and estimated contaminants data (Laird et al. 2005; Kumar et al. 2012). Methods based on back tracking of contaminants are developed, which reformulate the partial differential pipe expressions into a set of algebraic constraints (Laird et al. 2005). Mixed integer quadratic program formulation is applied to identify the number of likely injection locations (Laird et al. 2006). Particle backtracking algorithm to identify the water flow paths and travel times for each monitoring station, with a simple rule to determine if any node and time interval is safe, unsafe, or unknown (De Sanctis et al. 2009).

5 Research Needs

Notable progress has been made in understanding the change patterns of water quality parameters and the reactivity and transport modeling of contaminants. Several methodologies of water quality event detection are developed. Since it is the objective to detect the contamination event more quickly and accurately, further researches are needed and a suggested area of researches are listed below.

Research for More Contaminants Experiment results show the correlation of contaminants and water quality parameters (USEPA 2005a; Jeffrey Yang et al. 2009; Byer and Carlson 2005). However, the contaminants used in the experiments are limited, so a broader selection of contaminants is needed, especially organic chemicals. A comprehensive study about the correlation of contaminants and water quality parameters is needed.

Contaminant Identification When contamination event occurs, to identify the contaminant as quickly as possible is important. Automatic online contaminant identification method has been reported (Kroll and King 2006) which compare the deviation vector with patterns stored. However, a more comprehensive library for more contaminants is still needed. Models for reactivity and transport of contaminants (Jeffrey Yang et al. 2008) can be used for contaminant identification, which needs further research.

Data Pattern Extraction Though several event detection methods are proposed, we know a little about the

water quality data normal pattern. The normal pattern is very important for the better understanding of water quality, and is very helpful for event detection. So, extract pattern for daily water quality monitoring data is a meaningful research direction.

Test Using Real Event Several water quality event detection methodologies are developed and tested. However, simulated events are mostly used for performance test. But simulated event can not characterize the real contaminant event. So real contamination data is needed to test the event detection methods, and in-situ application of event detection system is also needed.

Detection Using UV-Vis Spectrum Ultraviolet-visible (UV-Vis) spectrum is mostly applied in water quality monitoring mostly for organic contaminant monitoring (Hou et al. 2013d; Youquan et al. 2010), based on the Beer's law and the fact that the chemicals in water strongly absorb UV radiation (Fogelman et al. 2006). The UV-Vis monitoring has the advantage that a sample can be analyzed with little sample preparation, an absence of chemicals, and relatively low operational cost (Fogelman et al. 2006). Some methodologies to use the spectrum directly for water quality event detection are developed (Langergraber et al. 2004). However, further researches are still needed for deeper application of the UV-Vis spectrum.

6 Conclusion

Contamination warning systems are set up to safeguard the drinking water. Event detection methodologies are proposed base on the fact that several water quality parameters change significantly when contaminants are introduced into water, and free chlorine is the most sensitive parameter in water network. Event detection methodologies consist of sensor data analysis, event detection method, and performance test. Usually, in event detection, current data pattern is compared with the normal to declare a outlier or event. Optimal sensor placement in drinking water network is helpful for quick contamination detection.

Several fields need further research, including a border range of contaminants, contaminants identification using water quality data, normal data

pattern extraction, and UV-Vis spectrum-based event detection.

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